Composition Order and Inverse Domains

Algebra 2 — Error Analysis

Directions: These problems are about two errors that look harmless on paper: composing in the wrong order, and accepting an inverse without checking its domain. Show the substitution you actually performed. On the find-and-fix problems (4 and 6), name the mistake in one sentence *and* give the corrected value.

1. Let $f(x) = 2x - 5$ and $g(x) = x^2 + 1$.
Find $(f \circ g)(3)$. Write down $g(3)$ first, then feed it to f.

Answer: __________________

2. Using the same $f(x) = 2x - 5$ and $g(x) = x^2 + 1$, find $(g \circ f)(3)$.
Then compare with your answer to problem 1 and state in one sentence what the comparison shows about the order of composition.

Answer: __________________

3. Let $f(x) = 3x + 7$. The inverse f^{-1} undoes f, so $f^{-1}(22)$ is the input that f sends to 22.
Solve $3x + 7 = 22$ to find $f^{-1}(22)$.

Answer: ____________

4. **Find and fix.** With $f(x) = 3x + 1$ and $g(x) = x - 6$, Nia was asked for $(f \circ g)(4)$. She wrote:

$$(f \circ g)(4) = f(4) - 6 = 13 - 6 = 7$$

Explain in one sentence which composition Nia actually computed, then compute $(f \circ g)(4)$ correctly.

Answer: ____________

5. Let $f(x) = 3x + 7$, whose inverse is $f^{-1}(x) = \dfrac{x-7}{3}$.
Show that $f\big(f^{-1}(x)\big)$ simplifies to x for every real x. Write each line of the simplification.

Answer: ________________

6. **Find and fix.** Let $h(x) = (x-3)^2$ with the domain restricted to $x \geq 3$. Sam claims the inverse is $h^{-1}(x) = 3 - \sqrt{x}$, and reports $h^{-1}(16) = -1$.
Test Sam's claim by checking whether -1 lies in the domain $x \geq 3$ of h. Then state the correct inverse rule and compute $h^{-1}(16)$.

Answer: ________________

7. Let $g(x) = x^2$ with *no* restriction on its domain, and suppose someone writes "$g^{-1}(x) = \sqrt{x}$".
Solve $x^2 = 9$ and list every real solution. Then explain in one sentence what your list proves about g having an inverse on all of $\mathbb{R}$.

Answer: __________________

8. Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$. Marcus checks $f(g(x)) = \sqrt{(x^2+1)-1} = \sqrt{x^2} = x$ and concludes that f and g are inverses for every real number x.
Explain why his conclusion holds only for $x \geq 0$, state which composition check he skipped, and give the domain of f and the domain of g that make the pair genuine inverses.